# Baseball data: Learning Activity 4

## Background

This activity uses the **baseball** dataset (available in the *SDAResources* package). The dataset contains statistics on **797 baseball players** from the rosters of all major league teams in November 2004 (Forman, 2004).

You will treat this file as a **population** and draw samples from it using stratified random sampling.

## Getting Started

Loading the dataset and the R packages. Visit the solutions for learning activity 3 on R introductions.

```
# Uncomment the install.packages("SDAResources") line to install the package.
# Comment it back after you install the package :)
# install.packages("SDAResources")
library(SDAResources)
library(dplyr)

data(baseball)
```

## Exercise 1: Stratified random sample

Take a stratified random sample of 150 players from the population, using proportional allocation with the different teams as strata. Save your sample as `baseball_strs` for use in future exercises. Don't forget to set your seed. I've included some code to get you started. NOTE this chunk won't run until you fill in the missing blanks.

```r
# Set seed for reproducibility
set.seed(123)

# Total sample size
n <- 150

# Total population size
N <- nrow(baseball)

# Step 1: Get stratum sizes (Nh = number of players per team)
prop_alloc <- baseball %>%
  group_by(team) %>%
  summarise(Nh = n())

# Step 2: Calculate proportional sample size for each stratum
prop_alloc <- prop_alloc %>%
  mutate(nh = round( Nh/ N * n))

# Step 3: Create an empty data frame to store the sample
baseball_strs <- data.frame()

# Step 4: Loop through each team and sample nh players
for (i in 1:nrow(prop_alloc)) {

  # Get the current team name
  current_team <- prop_alloc$team[i]

  # Get the sample size for this team
  current_nh <- prop_alloc$nh[i]

  # Get the Stratum size for this team
  current_Nh <- prop_alloc$Nh[i]

  # Filter the data to only include the current team
  team_data <- baseball %>%
    filter(team == current_team)

  # Take a random sample of current_nh players from this team
  team_sample <- team_data[sample(1:nrow(team_data), size = current_nh), ]

  # Append the stratum size and sample size so you have easy access for calculations later
  team_sample$nh = current_nh
```

```
  team_sample$Nh = current_Nh

  # Add the team sample to our growing stratified sample
  baseball_strs <- rbind(baseball_strs, team_sample)
}
```

```
head(baseball_strs)
```

```
# A tibble: 6 x 32
  team  leagueid player    salary pos   gplay gstart inning putout assist error
  <chr> <chr>    <chr>      <dbl> <chr> <dbl>  <dbl>  <dbl>  <dbl>  <dbl> <dbl>
1 ANA   AL       lackejo0  375000 P         2     32    595     15     23     0
2 ANA   AL       pauljo01  335000 C        46     16    504    134      9     1
3 ANA   AL       kennead0 2500000 2B      144    138   3675    255    388    12
4 ANA   AL       davanje0  375000 CF      108     27    743     75      1     0
5 ANA   AL       greggke0  301500 P         5      0    263      2      5     0
6 ARI   NL       mantema0 7000000 P        12      0     32      1      2     0
# i 21 more variables: dplay <dbl>, pball <dbl>, gbat <dbl>, atbat <dbl>,
#   run <dbl>, hit <dbl>, secbase <dbl>, thirdbase <dbl>, homerun <dbl>,
#   rbi <dbl>, stolenb <dbl>, csteal <dbl>, walk <dbl>, strikeout <dbl>,
#   intwalk <dbl>, hbp <dbl>, sacrhit <dbl>, sacrfly <dbl>, gidplay <dbl>,
#   nh <dbl>, Nh <int>
```

Describe how you selected the sample here: We first split the data into strata defined by teams. We then took an SRS of size 5 within each strata.

### Exercise 2: Inference for the population proportion

Estimate the population proportion of players who are pitchers and give a 95% confidence interval, using your stratified sample. Do this manually using the formulas from lecture/the textbook — do not rely on pre-built R functions except for `mean()`, `sd()`, `qnorm(0.975)`, or `qt(0.975, df = n - 1)`, which give you the 97.5th quantile from a standard normal distribution and a t-distribution with $n-1$ degrees of freedom, respectively.

```
## Include your code here

df_forcalc <- baseball_strs %>%
  group_by(team) %>%
  summarise(phat = mean(pos == "P"),
            nh = nh[1], # just grabs the first one (they are all the same for a specific tean
```

```r
            Nh = Nh[1]) # just grabs the first one (they are all the same for a specific tean

df_forcalc %>% head(5)
```

```
# A tibble: 5 x 4
  team   phat    nh    Nh
  <chr> <dbl> <dbl> <int>
1 ANA     0.4     5    26
2 ARI     0.8     5    28
3 ATL     0.6     5    28
4 BAL     0.2     5    25
5 BOS     0       5    27
```

```r
### Include your code here to calculate the confidence interval

phat_str = sum(df_forcalc$Nh/N*df_forcalc$phat)
variance_hat = sum((df_forcalc$Nh/N*df_forcalc$phat)^2*phat_str*(1-phat_str)/(df_forcalc$nh-

ci_lwr = phat_str - qt(0.975, df = 150-1)*sqrt(variance_hat)
ci_upr = phat_str + qt(0.975, df = 150-1)*sqrt(variance_hat)

ci_lwr
```

```
[1] 0.4429717
```

```r
ci_upr
```

```
[1] 0.5311814
```

Give the estimate and the 95% CI and interpret them here: The estimate is 0.487 and the CI is [0.44, 0.53]. We are 95% confident that the true proportion of pitchers is in this interval. We estimate the proportion of pitchers to be 0.487.

### Exercise 3: Inference for the population mean

Estimate the population mean of `logsal` and give a 95% CI, using your stratified sample with proportional allocation. Recall from learning activity 3 that `logsal` is the logarithm of `salary`. Assume the strata population variances are unknown here.

Do this manually using the formulas from lecture/the textbook — do not rely on pre-built R functions except for `mean()`, `sd()`, `qnorm(0.975)`, or `qt(0.975, df = n - 1)`, which give you the 97.5th quantile from a standard normal distribution and a t-distribution with $n - 1$ degrees of freedom, respectively.

```
# Include your code here

baseball_strs <- baseball_strs %>%
  mutate(logsal = log(salary))

df_forcalc <- baseball_strs %>%
  group_by(team) %>%
  summarise(ymean = mean(logsal),
            sd = sd(logsal),
            nh = nh[1], # just grabs the first one (they are all the same for a specific tean
            Nh = Nh[1]) # just grabs the first one (they are all the same for a specific tean

df_forcalc %>% head(5)
```

```
# A tibble: 5 x 5
  team  ymean    sd    nh    Nh
  <chr> <dbl> <dbl> <dbl> <int>
1 ANA    13.1 0.890     5    26
2 ARI    13.5 1.33      5    28
3 ATL    14.8 1.57      5    28
4 BAL    14.4 1.08      5    25
5 BOS    14.8 1.42      5    27
```

```
### Include your code here to calculate the confidence interval

ymean_str = sum(df_forcalc$Nh/N*df_forcalc$ymean)
variance_hat = sum((1-df_forcalc$nh/df_forcalc$Nh)*(df_forcalc$Nh/N)^2*df_forcalc$sd^2/df_fo

ci_lwr = ymean_str - qt(0.975, df = 150-1)*sqrt(variance_hat)
ci_upr = ymean_str + qt(0.975, df = 150-1)*sqrt(variance_hat)

ci_lwr
```

```
[1] 13.9106
```

```r
ci_upr
```

```
[1] 14.25844
```

Give the estimate and the 95% CI and interpret them here: We estimate the population mean for log salary to be 14.1. We are 95% confident that the true population mean for log salary is between 13.91 and 14.26.

### Exercise 4: Comparison to CI from Learning Activity 3

How do your estimates compare with those of Learning activity 3?

In learning activity 3, the 95% CI for log salary was [13.63, 13.97]. Note that this was obtained on a different sample and under a different sampling strategy (SRS). Both confidence intervals overlap with the interval being slightly higher under stratified sampling. The width of the confidence interval under stratified sampling is 0.35 vs 0.34 for SRS and therefore comparable. In these realizations, there was no benefit, in terms of precision, between the two sampling schemes. This could indicate that the mean log salary between each strata (team) are not too different from each other.

### Exercise 5: Justification for optimal allocation

Examine the population standard deviation in each stratum. I've included some code for you to fill out.

```r
## Use this for code

sh <- baseball %>%
  mutate(logsal = log(salary)) %>%
  group_by(team) %>%
  summarise(sd = sd(logsal))

sh
```

```
# A tibble: 30 x 2
   team     sd
   <chr> <dbl>
 1 ANA    1.41
 2 ARI    1.32
 3 ATL    1.38
```

```
 4 BAL    1.11
 5 BOS    1.27
 6 CHA    1.28
 7 CHN    1.21
 8 CIN    1.15
 9 CLE    1.00
10 COL    1.22
# i 20 more rows
```

Do you think optimal allocation would be worthwhile for this problem? Justify your response.

The variabiltiy of log income appears to change accross strata. For example, CLE has a standard deviation of 1 and ANA has a standard deviation of 1.4. It might be better to re-allocate observations to to strata with larger variability in order to obtain more precise estimate of the stratum specific means.

### Exercise 6: Computing optimal allocations

Use the population standard deviations from Exercise 5 to determine the optimal allocation for a sample of size 150.

```r
opt_alloc <- prop_alloc

# Include your code here

# Adding the stratas population standard deviations
opt_alloc <- opt_alloc %>%
  left_join(sh, by = "team")

# Calculate optimal allocation sample size for each stratum
opt_alloc <- opt_alloc %>%
  mutate(nh_opt = round( Nh*sd/ sum(Nh*sd)*150))   # FILL IN

head(opt_alloc)
```

```
# A tibble: 6 x 5
  team     Nh    nh    sd nh_opt
  <chr> <int> <dbl> <dbl>  <dbl>
1 ANA      26     5  1.41      6
2 ARI      28     5  1.32      6
3 ATL      28     5  1.38      6
```

```
4 BAL       25      5  1.11       4
5 BOS       27      5  1.27       5
6 CHA       26      5  1.28       5
```

How much does the optimal allocation differ from the proportional allocation? Do you observe any patterns in how observations were re-allocated?

More observations were allocated to teams/strata that had higher variability in log income values. Optimal allocation allocates observations proportional to both the strata size and the variability of log income within each strata. Compare CHA and ANA for example. Both these teams were the size size but ANA had more variability in the log income. We can see that rather than allocating 5 observations to each team, 6 observations were allocated to ANA (with the larger variability) and 5 to CHA.

### Exercise 7: Stratified random sample using optimal allocation

Take a stratified random sample of 150 players from the population, using optimal allocation with the different teams as strata. Save your sample as `baseball_strs_opt` for use in future exercises. Don't forget to set your seed. I've included some code to get you started. NOTE this chunk won't run until you fill in the missing blanks.

```r
# Set seed for reproducibility
set.seed(123)

# Step 3: Create an empty data frame to store the sample
baseball_strs_opt <- data.frame()

# Step 4: Loop through each team and sample nh players
for (i in 1:nrow(opt_alloc)) {

  # Get the current team name
  current_team <- opt_alloc$team[i]

  # Get the sample size for this team
  current_nh <- opt_alloc$nh_opt[i]

  # Get the Stratum size for this team
  current_Nh <- opt_alloc$Nh[i]

  # Filter the data to only include the current team
  team_data <- baseball %>%
    filter(team == current_team)
```

```
  # Take a random sample of current_nh players from this team
  team_sample <- team_data[sample(1:nrow(team_data), size = current_nh), ]

  # Append the stratum size and sample size so you have easy access for calculations later
  team_sample$nh = current_nh
  team_sample$Nh = current_Nh

  # Add the team sample to our growing stratified sample
  baseball_strs_opt <- rbind(baseball_strs_opt, team_sample)
}
```

```
head(baseball_strs_opt)
```

```
# A tibble: 6 x 32
  team  leagueid player    salary pos   gplay gstart inning putout assist error
  <chr> <chr>    <chr>      <dbl> <chr> <dbl>  <dbl>  <dbl>  <dbl>  <dbl> <dbl>
1 ANA   AL       lackejo0  375000 P         2     32    595     15     23     0
2 ANA   AL       pauljo01  335000 C        46     16    504    134      9     1
3 ANA   AL       kennead0 2500000 2B      144    138   3675    255    388    12
4 ANA   AL       davanje0  375000 CF      108     27    743     75      1     0
5 ANA   AL       greggke0  301500 P         5      0    263      2      5     0
6 ANA   AL       ortizra0 3266667 P         1     14    384      6     13     2
# i 21 more variables: dplay <dbl>, pball <dbl>, gbat <dbl>, atbat <dbl>,
#   run <dbl>, hit <dbl>, secbase <dbl>, thirdbase <dbl>, homerun <dbl>,
#   rbi <dbl>, stolenb <dbl>, csteal <dbl>, walk <dbl>, strikeout <dbl>,
#   intwalk <dbl>, hbp <dbl>, sacrhit <dbl>, sacrfly <dbl>, gidplay <dbl>,
#   nh <dbl>, Nh <int>
```

### Exercise 7: Inference for the population mean under optimal allocation

Estimate the population mean of `logsal` and give a 95% CI, using your stratified sample with optimal allocation. Assume strata population variances are unknown.

Do this manually using the formulas from lecture/the textbook — do not rely on pre-built R functions except for `mean()`, `sd()`, `qnorm(0.975)`, or `qt(0.975, df = n - 1)`, which give you the 97.5th quantile from a standard normal distribution and a t-distribution with $n-1$ degrees of freedom, respectively.

```
# Include your code here

baseball_strs_opt <- baseball_strs_opt %>%
```

```r
  mutate(logsal = log(salary))

df_forcalc <- baseball_strs_opt %>%
  group_by(team) %>%
  summarise(ymean = mean(logsal),
            sd = sd(logsal),
            nh = nh[1], # just grabs the first one (they are all the same for a specific tean
            Nh = Nh[1]) # just grabs the first one (they are all the same for a specific tean

df_forcalc %>% head(5)
```

```
# A tibble: 5 x 5
  team  ymean    sd    nh    Nh
  <chr> <dbl> <dbl> <dbl> <int>
1 ANA    13.5 1.10      6    26
2 ARI    13.2 0.874     6    28
3 ATL    14.7 1.21      6    28
4 BAL    14.2 0.938     4    25
5 BOS    14.6 1.18      5    27
```

```r
### Include your code here to calculate the confidence interval

ymean_str = sum(df_forcalc$Nh/N*df_forcalc$ymean)
variance_hat = sum((1-df_forcalc$nh/df_forcalc$Nh)*(df_forcalc$Nh/N)^2*df_forcalc$sd^2/df_fo

ci_lwr = ymean_str - qt(0.975, df = 150-1)*sqrt(variance_hat)
ci_upr = ymean_str + qt(0.975, df = 150-1)*sqrt(variance_hat)

ci_lwr
```

```
[1] 13.65534
```

```r
ci_upr
```

```
[1] 13.99882
```

Give the estimate and the 95% CI and interpret them here. How do they compare with the confidence interval obtained in exercise 3?

The 95% CI is 13.65 to 13.40. We are 95% confident the population mean for log salary is within this interval. The width of this confidence interval is 0.343 and the width of the previous confidence interval using proportional allocation was 0.348. So the width of the confidence interval decreased slightly with optimal allocation.